Beyond Demographic-Free Auditing: Measuring and Mitigating Hidden-Group Bias with Few Labels

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***Abstract*—Fairness audits and interventions usually require a protected attribute that practitioners often cannot collect for technical or legal reasons. We study what can be done when demographic data are unavailable except for a small labelled sample. On 8 benchmark dataset-attribute pairs and 10 American Community Survey (ACS) targets, under a pre-registered protocol that never tunes on demographic labels and scores every model at a fixed selection rate, we separate three tasks: finding, measuring and fixing a hidden-group disparity. We first show that, within a true label, a group's mean score gap equals the score's standard deviation times its correlation with the group, divided by the square root of the group's balance. This identity explains why a model trained without the protected attribute shows larger gaps on its own input features than on the hidden group: in 660 demographic-free audits, the protected group never ranked first. Second, about 100 labelled rows suffice to measure the hidden equalized-odds gap; a proxy-weighted estimator is nearly unbiased and 2 to 4 times more accurate than the labelled rows alone, on all 18 targets. Third, mitigation is conditional. On the benchmarks, a 100-label proxy removed 14.5% of the gap, against at most 7.3% for demographic-free methods, but on ACS the unguarded proxy did not reliably help, and its benefit tracked the size of the baseline gap. A simple guard that corrects only when the estimated gap is at least 0.08 cut harmful corrections from 26% to 7% on ACS and from 29% to 11% in a pre-registered replication on five new states.**

***Index Terms*—algorithmic fairness, fairness without demographics, proxy attributes, fairness auditing, equalized odds, pre-registration**

I. Introduction

Most group-fairness methods assume access to a protected attribute A, such as race or sex, for every training example [1]–[3]. In practice this information is often missing: collecting it is expensive or legally restricted, people may decline to disclose it, or it was never recorded.

Two broad strategies address this problem. *Demographic-free* methods try to improve fairness without any demographic labels, either by giving more weight to examples that are hard to fit [4]–[7] or by penalising features suspected to be linked to A [8]. *Limited-demographics* methods use a small labelled sample to train a proxy for A and then use the proxy to audit the model or to mitigate bias [9], [10].

We ask a practical question: *with few or no demographic labels, what can be measured, and when can a disparity be fixed safely?* We separate three tasks that are often treated as one: finding the disadvantaged group, measuring how large the disadvantage is, and reducing it. A method that succeeds at one task need not succeed at the others.

Our evaluation is deliberately conservative. Demographic labels are never used to tune hyper-parameters, and every model is scored at the same selection rate, so no model can look fair simply by predicting the favourable outcome for everyone. The confirmatory experiments follow a pre-registration written after pilots on separate seeds [11].

Our contributions are:

- **Why demographic-free audits miss the protected group.** An exact identity links a group's mean score gap to its correlation with the score (Proposition 1). A model trained without A is associated with A only through proxies, so the features it uses most dominate a gap-ranking audit; A ranked first in 0 of 660 audits.
- **Measurement from about 100 labels.** A proxy-weighted estimator recovers the hidden equalized-odds gap almost without bias and beat the labelled-rows estimate on all 18 targets, including a pre-registered replication.
- **A conditional answer for mitigation.** Proxy correction helped on the benchmarks but needed a guard on ACS: its benefit tracked the baseline gap (Spearman correlation 0.85 over 13 targets), and a simple gap guard held in a held-out test and in a pre-registered replication.
- **Negative results.** Selective labelling, tuning on the labelled rows and four demographic-free methods did not give reliable gains.

II. Related Work

Group fairness and mitigation with observed attributes. Equalized odds and equal opportunity require similar error rates across groups conditional on the true label [1]; we use equalized odds throughout. Mitigation methods are commonly grouped into pre-processing, such as reweighing and massaging of the training data [3]; in-processing, such as the reductions approach that turns a fairness constraint into a sequence of cost-sensitive problems [2]; and post-processing of scores or thresholds [1]. All of these assume that A is observed for every training example, which is the assumption this paper relaxes. Our learner is a simple in-processing method with a soft equalized-odds penalty, chosen so that every arm differs only in the group signal it receives, and we reuse the Kamiran-Calders weights inside one of our baselines [3].

Fairness without demographics. Distributionally robust optimization minimises the loss of the worst-off portion of the data without group labels, either over repeated rounds of loss minimisation [4] or with efficient large-scale CVaR objectives

[7]. ARL trains an adversary that assigns higher weight to examples the learner finds hard [5]; JTT upweights examples misclassified by an initial model [6]; and clustering approaches construct pseudo-groups from learned representations [12]. FairRF instead penalises correlations between predictions and user-specified related features [8]. These methods share a premise: the people a model serves worst coincide with a protected group. Kenfack et al. report that they do not necessarily improve demographic parity or equalized odds for the true attribute [9], and hyperparameter selection is a further difficulty when A is unavailable for validation [13]. Our score-correlation identity gives a direct reason why the premise can fail: the worst-off regions are defined by the features the model relies on, not by a hidden attribute. We reimplement ARL, CVaR-DRO, JTT and cluster reweighting under one learner so that their comparison with proxy methods is not confounded by architecture or tuning.

Limited demographics and proxies. A second line of work assumes that A is known for a subset of individuals. Kenfack et al. train an uncertainty-aware attribute classifier from a labelled subset representing 3% to 20% of the data and apply fairness constraints only to rows whose predicted attribute is confident [9]. Awasthi et al. study fairness evaluation when the attribute predictor is noisy or incomplete, and show that the error of the predictor can bias the estimated disparity in either direction [10]. We work with much smaller labelling budgets, from 25 to 400 rows, keep demographic information out of model tuning, and identify the baseline gap rather than proxy accuracy as the quantity that predicts when proxy-based correction makes a model less fair.

Measurement of disparity with an unobserved attribute. Chen et al. analyse proxy-weighted disparity estimators under unawareness and characterise when they over- or under-state the true disparity [14]. Kallus et al. derive partial-identification bounds on disparity by combining the main dataset with an auxiliary dataset that contains the protected attribute [15]. Elzayn et al. estimate and enforce conventional fairness metrics using probabilistic protected features [16]. Our measurement results sit within this line of work while focusing on very small label budgets, comparing directly against the naive labelled-rows estimate that a practitioner would compute first, and evaluating on two data families.

Benchmarks and evaluation practice. We use standard tabular fairness datasets collected by Le Quy et al. [17]. Because older benchmarks such as Adult have been criticised for their age and construction, we also use the ACS Folktables tasks [18], and we test our main decision rule on ACS states that were not used to design it. Following practice from the experimental sciences, our confirmatory analyses were pre-registered before they were run, with deviations logged [11].

Positioning. Prior work has typically studied either demographic-free or limited-demographics methods, and either measurement or mitigation. We evaluate both families under one learner, one tuning rule and one fixed selection rate; separate the tasks of finding, measuring and fixing a disparity; and connect the outcome of each task to a single quantity, namely how strongly the model's score is associated with the group.

III. Setup and Methods

Fig. 1 gives an overview of the full study design.

A. Notation and metric

Each example has features X, a binary label Y and a binary protected attribute A that is *never* used as a model input. A classifier produces a score s and a decision $\hat{Y}$, with the threshold chosen so that the selection rate matches the training base rate. Writing TPR_a and FPR_a for the true- and false-positive rates in group $A = a$, the equalized-odds gap is

$$\mathrm{EO} = \max\big(\,|\mathrm{TPR}_1 - \mathrm{TPR}_0|,\ |\mathrm{FPR}_1 - \mathrm{FPR}_0|\,\big). \quad (1)$$

We report the relative reduction against the unconstrained model's gap EO_0, and absolute changes when EO_0 is small.

B. Learner and group signals

All mitigation arms use the same learner, logistic regression with predicted probabilities p_i and the loss

$$\mathcal{L} = \frac{1}{n}\sum_i \ell(p_i, y_i) + \lambda\|w\|^2 + \mu \sum_{y\in\{0,1\}} \big(\bar{p}_{1y} - \bar{p}_{0y}\big)^2, \quad (2)$$

where $\bar{p}_{ay}$ is the mean of p_i over examples with label y in group a, weighted by a group signal $g_i \in [0,1]$ (and by $1-g_i$ for $a=0$).

The arms differ only in how g is obtained. The *oracle* uses the true attribute and serves only as a reference. The *proxy* uses $\hat{q}_i$, the probability of $A=1$ predicted by a logistic regression fitted on n labelled rows. We also test a FairDSR-style variant that uses hard proxy labels on the more confident half of rows [9], and a FairRF-style variant that penalises the five features most correlated with A in the labelled rows [8].

C. Demographic-free baselines

ARL with a linear learner and adversary [5], CVaR-DRO [7], JTT [6], and cluster reweighting each receive a sweep over their strength parameter. Cluster reweighting uses k-means pseudo-groups with Kamiran-Calders weights [3].

D. Choosing which rows to label

Demographics are revealed only for rows selected from a labelling pool. The pilot compared random, outcome-stratified, diverse (k-means centres), uncertainty-based and minority-seeking selection; the confirmatory runs keep random and outcome-stratified selection. We also consider a disclosure-bias condition in which members of the protected group disclose at one third the rate of others.

E. Tuning without demographics

For each arm, we keep the strongest setting whose pool AUC stays within 0.01 of the unconstrained model (rule T1). If no setting satisfies this constraint, we keep the unconstrained model. A label-informed rule (T2) instead selects, within the same AUC budget, the setting with the smallest EO gap on the labelled rows.

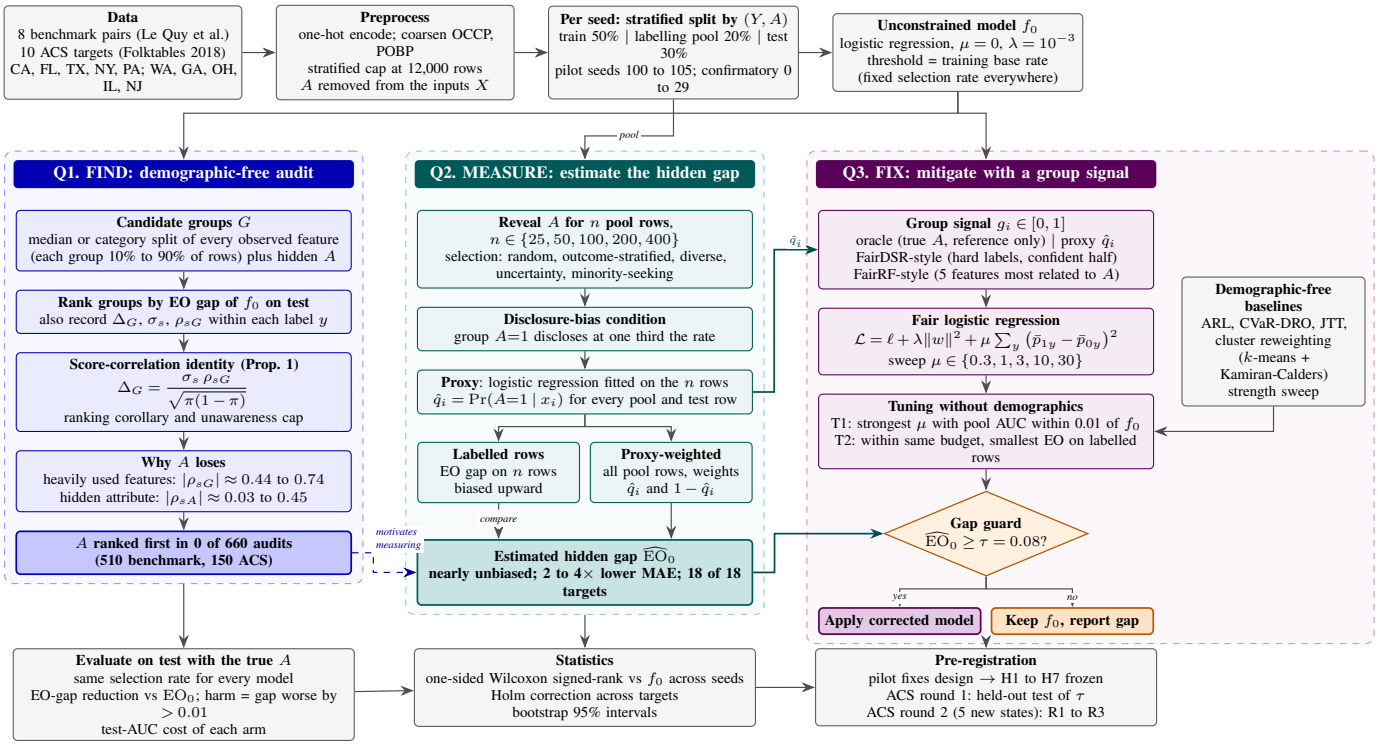


Fig. 1. End-to-end study design. The hidden attribute A is removed from every model input and used only for the labelled sample and for final grading. Each seed feeds three questions: Q1 asks whether a demographic-free audit can *find* the disadvantaged group (Section IV); Q2 asks whether n revealed labels and a proxy can *measure* the hidden equalized-odds gap; Q3 asks whether the same signal can *fix* it, comparing proxy, oracle, FairDSR- and FairRF-style variants and four demographic-free baselines under one learner and one tuning rule. The measured gap from Q2 gates the correction in Q3 through the gap guard. All arms are graded on the test split with the true A at a fixed selection rate.

F. Measuring the hidden gap

Both estimators use the unconstrained model's decisions on the pool. The *labelled-rows* estimator computes the EO gap using only the n labelled rows. The *proxy-weighted* estimator uses every pool row, weighting it by $\hat{q}_i$ for group 1 and by $1-\hat{q}_i$ for group 0; for example,

$$\widehat{\mathrm{TPR}}_1 = \frac{\sum_{i:\, y_i=1} \hat{q}_i\, \hat{y}_i}{\sum_{i:\, y_i=1} \hat{q}_i}, \tag{3}$$

and similarly for the other three rates.

G. Gap guard

Proxy correction is applied only if the proxy-weighted estimate of EO_0 is at least τ; otherwise, the unconstrained model is kept. The threshold $\tau = 0.08$ was chosen on the benchmark data and fixed before any ACS run.

IV. Why Demographic-Free Audits Miss the Protected Group

A demographic-free audit ranks candidate groups, usually defined by observed features, according to their error or score gap and flags the largest one. The following identity shows what this ranking actually measures. Fix a label y, and let π be the share of group G among examples with that label.

Proposition 1 (Score-correlation identity)**.** *For any score s and binary group G with $0 < \pi < 1$,*

$$\Delta_G = \frac{\sigma_s\, \rho_{sG}}{\sqrt{\pi(1-\pi)}}, \tag{4}$$

where Δ_G is the difference in mean score between $G = 1$ and $G = 0$, σ_s is the standard deviation of s, and ρ_{sG} is the correlation of s with the indicator of G, all within label y.

Proof. Within label y, the covariance of s and G equals $\pi(1-\pi)\Delta_G$, and the standard deviation of G is $\sqrt{\pi(1-\pi)}$. Dividing the covariance by $\sigma_s\sqrt{\pi(1-\pi)}$ gives ρ_{sG}. □

Corollary 1 (Ranking)**.** *The protected group outranks a feature group G only if*

$$\frac{|\rho_{sA}|}{\sqrt{\pi_A(1-\pi_A)}} > \frac{|\rho_{sG}|}{\sqrt{\pi_G(1-\pi_G)}}. \tag{5}$$

If the score rises with one roughly Gaussian feature, the median split of that feature has $|\rho_{sG}| \approx 0.80$, so a group making up 15% of rows needs $|\rho_{sA}| > 0.57$.

Corollary 2 (Unawareness cap)**.** *If the model does not take A as input, $|\rho_{sA}|$ is at most the maximal correlation between A and any function of the features. The protected group can top the audit only when the features nearly reveal it.*

The practical implication is that the audit is largely measuring how strongly the model's score depends on each candidate

TABLE I

DEMOGRAPHIC-FREE AUDIT OF AN UNCONSTRAINED MODEL: RANK OF THE HIDDEN ATTRIBUTE AMONG FEATURE-DEFINED GROUPS (1 = LARGEST EO GAP) AND SCORE CORRELATIONS. MEDIANS OR MEANS OVER 30 SEEDS.

Target	EO gap of A	Rank of A	$\lvert\rho_{sA}\rvert$	max $\lvert\rho_{sG}\rvert$
Law · race	0.39	6 of 8	0.45	0.74
COMPAS · race	0.25	5 of 5	0.28	0.53
German · age	0.17	16.5 of 37	0.14	0.54
Credit · age	0.17	12 of 22	0.14	0.74
Adult · sex	0.16	14 of 18	0.24	0.67
COMPAS · sex	0.14	5 of 5	0.13	0.55
OULAD · disability	0.11	16 of 24	0.09	0.51
Dutch · sex	0.05	15 of 19	0.04	0.61
ACS CA · race	0.06	19	0.09	0.44
ACS FL · race	0.11	16	0.10	0.49
ACS TX · race	0.06	18	0.07	0.49
ACS NY · sex	0.02	22	0.03	0.44
ACS PA · sex	0.06	17	0.07	0.70
A ranked first: 0 of 510 benchmark audits (17 pairs); 0 of 150 ACS audits.				

ACS audits had 19 to 23 candidate groups, including A. Nine further benchmark pairs (Bank, Diabetes and the second attribute of Adult, German, Law, Credit and OULAD) are included in the 510 audits.

TABLE II

TARGETS. FAVOURABLE LABEL IN PARENTHESES; A=1 DENOTES THE GROUP LISTED.

Source	Label	Hidden attribute (A=1)	Rows
Adult	income >50K	sex (female)	12,000
COMPAS	no 2-yr recidivism	race (Afr.-Am.); sex (female)	5,278
Law School	passes bar	race (non-white)	12,000
Credit (Taiwan)	no default	age <25	12,000
OULAD	passes course	disability	12,000
Dutch census	high occupation	sex (female)	12,001
German credit	good credit	age <25	1,000
ACS round 1	income / employment	CA, FL, TX race; NY, PA sex	12,000
ACS round 2	income / employment	WA, GA, OH race; IL, NJ sex	12,000

group indicator. Features that the model uses heavily therefore tend to produce large gaps. A hidden attribute can rank highly only when the observed features provide a strong proxy for it.

Table I confirms this pattern on both data families. The identity held to within 1.2×10^{-14} numerically, and on the benchmarks the ranking by mean score gap agreed with the ranking by thresholded EO gap (Spearman 0.85).

V. EXPERIMENTAL PROTOCOL

Data. Table II lists the targets. Benchmarks come from the collection of Le Quy et al. [17]; ACS data come from Folktables (2018, one-year person survey) [18], with occupation coarsened to major groups and place of birth to US-born or abroad region. Datasets above 12,000 rows were subsampled (stratified) to 12,000.

Splits and seeds. Each seed splits the data into training (50%), a labelling pool (20%) and test (30%), stratified by label and attribute. Label budgets are $n \in \{25, 50, 100, 200, 400\}$, capped at half the pool. Penalty strengths are $\mu \in \{0.3, 1, 3, 10, 30\}$ with $\lambda = 10^{-3}$.

Pre-registration. A pilot on seeds 100 to 105 compared the five labelling strategies and fixed the design; seven hypotheses (H1 to H7) were then frozen before confirmatory seeds 0 to 29. Two deviations were logged: extra backfire pairs used 10 seeds instead of 30, and some runs were split across processes. ACS round 1 tested the frozen guard threshold out of sample; ACS round 2 (five new states) was pre-registered with three hypotheses (R1 to R3) before it was run. Tests are one-sided Wilcoxon signed-rank tests across seeds against the unconstrained model, Holm-corrected across targets, with bootstrap 95% intervals.

TABLE III

ESTIMATING THE UNCONSTRAINED MODEL'S HIDDEN EO GAP FROM n LABELLED ROWS: MEAN ABSOLUTE ERROR (MAE) AND BIAS. L = LABELLED ROWS ONLY; P = PROXY-WEIGHTED. 30 SEEDS.

	Benchmarks (8 targets)				ACS round 1 (5 targets)			
n	MAE L	MAE P	Bias L	Bias P	MAE L	MAE P	Bias L	Bias P
25	0.284	0.093	+0.234	−0.001	0.241	0.059	+0.235	+0.037
50	0.226	0.097	+0.174	−0.004	0.175	0.047	+0.173	+0.023
100	0.163	0.072	+0.128	0.000	0.117	0.041	+0.111	+0.013
200	0.107	0.061	+0.065	−0.007	0.076	0.037	+0.062	+0.009
400	0.078	0.053	+0.038	−0.002	0.053	0.030	+0.034	−0.001

VI. RESULTS

A. Measurement: about 100 labels suffice

Table III compares the two estimators of the unconstrained model's hidden gap. The labelled-rows estimate is strongly biased upward at small n. The reason is that taking the absolute value of a noisy difference turns sampling variation into apparent unfairness. The proxy-weighted estimate is nearly unbiased and 2 to 4 times more accurate at $n \leq 100$.

The same pattern appears in the pre-registered ACS round 2. At $n = 100$, the proxy-weighted estimator was more accurate on all 5 new targets, with mean absolute error ranging from 0.03 to 0.06, compared with 0.08 to 0.16 for the labelled-rows estimate (R1 supported).

B. Mitigation on the benchmarks

Fig. 2 shows mean EO-gap reduction against the label budget on the 7 benchmark targets that allow every budget. With 100 random labels, the proxy reduced the gap by 15.5% [12.6, 18.2], compared with 6.9% for the best demographic-free method, cluster reweighting, and 27.0% for the oracle. ARL, CVaR-DRO and JTT remained between −2.5% and 1.6%.

Table IV gives the per-target results. The proxy matched the oracle on Adult and Law, recovered roughly half to four fifths of the oracle improvement on Credit and COMPAS, but struggled with OULAD, where the proxy was weak, and Dutch, where it actually reversed the gap. After Holm correction, the 100-label proxy was significant on 4 of 8 targets, the 400-label proxy on 6 of 7, the oracle on 7 of 8 and cluster reweighting on 3 of 8. Because the AUC budget is enforced on the pool, the realised test-AUC cost averaged at most 1.3 points for any arm and target.

C. ACS: the gap decides, and the guard holds

On ACS round 1, the unguarded proxy did not reproduce the benchmark result. At $n = 100$, it averaged −4.1% reduction

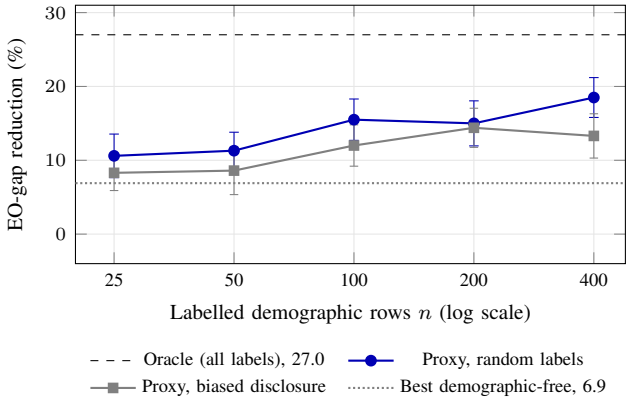


Fig. 2. Mean EO-gap reduction over 7 benchmark targets (German excluded: its pool cannot supply 200 or more labels), tuned without demographics within a 1-point AUC budget. Error bars: 95% bootstrap intervals over 30 seeds. ARL 1.3, CVaR-DRO 1.6, JTT −2.5.

TABLE IV
BENCHMARK EO-GAP REDUCTION (%) AT A 1-POINT AUC BUDGET, 30 SEEDS. CR = CLUSTER REWEIGHTING (BEST DEMOGRAPHIC-FREE METHOD).

Target	EO_0	CR	Proxy 100 [95% CI]	Proxy 400	Oracle
Law · race	0.39	1	30 [25, 35]	31	32
COMPAS · race	0.25	4	9 [6, 13]	11	11
German · age	0.17	10	8 [−4, 16]	n/a	2
Credit · age	0.17	31	20 [11, 27]	36	34
Adult · sex	0.16	28	48 [39, 55]	56	51
COMPAS · sex	0.14	3	5 [−1, 11]	12	11
OULAD · disab.	0.11	−4	2 [−4, 8]	10	32
Dutch · sex	0.05	−16	−5 [−17, 7]	−26	19
Mean (8)		7.3	14.5		23.9

compared with 8.1% for cluster reweighting, while the oracle still removed 31% to 58% of the gap on the four targets with $EO_0 \geq 0.05$.

The ACS proxies were uniformly weak, with proxy AUC between 0.58 and 0.70 at 100 to 400 labels. Proxy strength alone, however, did not explain the failures. NY · sex had one of the stronger proxies but produced the worst result. The baseline gap was much more informative. Across the 13 benchmark and round-1 targets (12 at 400 labels), EO_0 predicted the absolute benefit of both the 100-label and 400-label proxies, with Spearman correlation 0.85 in each case. When $EO_0 < 0.05$, correction made the gap worse in 38% of runs at 100 labels and 60% at 400 labels, compared with 15% and 5% for larger baseline gaps.

Table V reports the gap guard on all 10 ACS targets. In round 1, which served as a held-out test of the frozen threshold, the guard reduced the harm rate from 26% to 7% at 100 labels and from 18% to 1% at 400 labels. Mean reduction consequently increased from −4.1% to 1.6% at 100 labels and from 1.3% to 9.7% at 400 labels. In the pre-registered round 2, harm fell from 29% to 11% at 100 labels and from 23% to 5% at 400 labels (R2 supported), while the guarded 400-label proxy reduced the gap by 10.8% on targets with $EO_0 \geq 0.05$ (R3 supported).

TABLE V
ACS, 400 LABELS: EO-GAP REDUCTION (%) AND HARM RATE (GAP WORSE BY > 0.01) WITHOUT (U) AND WITH (G) THE 0.08 GAP GUARD. 30 SEEDS PER TARGET.

	Target	EO_0	Red. U	Red. G	Harm U	Harm G
Round 1	FL · race	0.108	39.2	31.9	0.00	0.00
	TX · race	0.057	26.1	13.0	0.07	0.00
	CA · race	0.056	36.4	11.4	0.00	0.00
	PA · sex	0.055	−0.6	0.0	0.20	0.00
	NY · sex	0.022	−94.4	−8.1	0.63	0.03
Round 2	NJ · sex	0.080	0.1	0.0	0.10	0.00
	GA · race	0.061	44.5	26.7	0.07	0.00
	WA · race	0.058	21.9	5.6	0.13	0.03
	OH · race	0.044	5.7	−0.2	0.17	0.03
	IL · sex	0.027	−96.4	−28.4	0.67	0.17

Pooled harm rate, unguarded to guarded: round 1, 0.26 to 0.07 (100 labels) and 0.18 to 0.01 (400 labels); round 2, 0.29 to 0.11 and 0.23 to 0.05.

The guard is not free. It trades some potential improvement for safety: GA falls from 44.5% to 26.7%, while CA falls from 36.4% to 11.4%. It also does not completely eliminate harm when the baseline gap is close to zero, as seen for IL · sex. Finally, a large gap is not sufficient on its own. NJ · sex starts with $EO_0 = 0.08$ but shows essentially no improvement.

D. Pre-registered outcomes and negative results

Table VI summarizes the confirmatory hypotheses. Four of seven benchmark hypotheses and all three round-2 hypotheses were supported.

Several negative findings matter as much. First, choosing which rows to label did not provide a clear advantage. In the pilot, uncertainty, minority-seeking, and diverse selection averaged 9.7% to 11.4% reduction, compared with 13.8% for random selection. Second, tuning on the labelled rows (T2) was 3.7 to 5.3 points worse than demographic-free tuning on the benchmarks, but 2.7 to 6.8 points better on ACS. Neither rule therefore dominated across datasets. Third, a pre-registered backfire diagnostic comparing the direction of the true-group and proxy-group gaps predicted harm only weakly (AUROC 0.58). This result motivated the simpler gap guard.

Disclosure bias consistently reduced the benefit, by 3.0 to 5.1 points on the benchmarks and 7.4 to 13.7 points on ACS. A neural-network check was inconclusive because even the oracle failed to improve under the chosen penalty grid.

VII. DISCUSSION AND LIMITATIONS

Implications. For auditors, a modest demographic survey of about 100 people combined with a population-wide proxy is enough to estimate equalized-odds gaps within a few points. Searching for the disadvantaged group without demographic labels, by contrast, tends to find the model's own features rather than the hidden protected group.

For practitioners, the key takeaway is to measure the gap before trying to correct it. When the baseline disparity is large, proxy correction can bring a meaningful improvement. When the estimated gap is small, correction with a weak proxy mostly

TABLE VI
CONFIRMATORY HYPOTHESES AND OUTCOMES.

Hypothesis	Outcome
H1: 100-label proxy beats every demographic-free method (benchmarks)	Supported (14.5 vs $\leq$7.3)
H2: 400-label proxy reaches 75% of oracle	Not supported (68%)
H3: stratified selection within 3 points of random	Supported
H4: disclosure bias lowers the benefit	Supported
H5: T2 not better than T1 at $n \leq 100$	Supported
H6: direction diagnostic predicts harm (AUROC $>$0.70)	Not supported (0.58)
H7: neural proxy beats ARL and JTT	Inconclusive
R1: proxy-weighted estimate better on $\geq$4 of 5 new targets	Supported (5 of 5)
R2: gap guard lowers harm at 100 and 400 labels	Supported
R3: guarded proxy positive where $\mathrm{EO}_0 \geq 0.05$	Supported (10.8%)

adds noise and can even reverse the disparity, so the measured gap is a useful gate for deciding whether to intervene.

Limitations. The protected attributes in our experiments are binary, so intersectional and multi-group settings remain untested. The proxies and primary learner are linear, and our neural-network check was inconclusive. ARL, CVaR-DRO, JTT and the FairDSR- and FairRF-style variants are our reimplementations rather than the authors' code.

The mitigation objective is equalized odds; methods targeting other fairness criteria may behave differently. We use a fixed selection-rate threshold to avoid any degenerate solutions, and a 0.5 threshold changed several baselines substantially in the pilot. Large datasets were subsampled to 12,000 rows, which may limit how well the results transfer to much larger populations. Finally, although the guard threshold was evaluated out of sample twice, it was originally selected using the benchmark data.

VIII. CONCLUSION

Without demographic data, a disparity audit tends to identify the model's own features rather than the protected group. An exact score-correlation identity explains why, and the pattern is supported by all 660 audits in our evaluation. With about 100 labelled rows, the hidden gap can instead be estimated accurately on both benchmark and census data.

Using those labels to reduce the disparity is more conditional. Proxy correction works when the baseline gap is sufficiently large, but can make small gaps worse. A simple pre-registered guard substantially reduces this risk.

In general, the results suggest treating a small demographic sample primarily as a measurement tool. Once the hidden disparity has been estimated, proxy-based intervention can be considered, but it should be gated by the size of the measured gap rather than applied automatically.

DATA AND CODE AVAILABILITY

Code, the pre-registration with logged deviations, raw per-run results and analysis scripts are available at https://github.com/tanistheta/Bias_detection. The ACS data are not redistributed; they are retrieved from the 2018 one-year person survey via Folktables [18]. An interactive demonstration that reruns the benchmark experiments for a chosen dataset, seed and label budget is available at https://hidden-group-bias.streamlit.app.

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